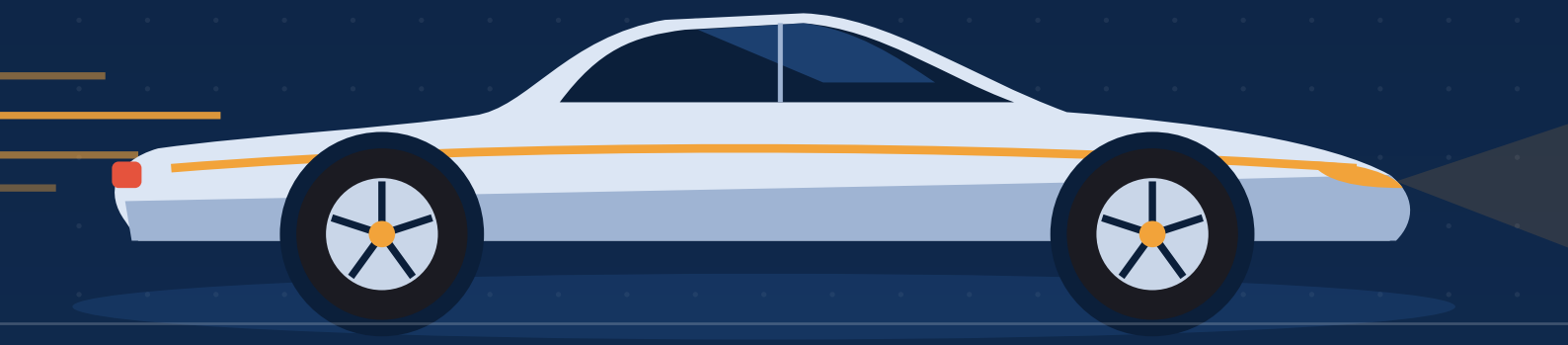
AIML | DATA ANALYTICS PROJECT

Exploratory Data Analysis on a Cars Dataset

Used-vehicle resale prices, depreciation and the factors that drive them



301	9	7	2003-18
LISTINGS IN DATASET	VARIABLES	CHARTS	MODEL YEARS

SUBMITTED BY
Bhavana HM

MENTOR
Sruthi Tarimana

COURSE
Artificial Intelligence and Machine Learning (AIML)

COLLEGE
GM Institute of Technology, Davangere

OPEN PROJECT NOTEBOOK (COLAB) >

01 Project Overview & Objective

Introduction

PROJECT TITLE	**Exploratory Data Analysis on a Cars Dataset**
STUDENT	**Bhavana HM**
MENTOR	**Sruthi Tarimana**
COURSE	Artificial Intelligence and Machine Learning (AIML)
COLLEGE	GM Institute of Technology, Davangere
NOTEBOOK	Open the project notebook in Google Colab

Objective

Used-vehicle prices are shaped by many factors at once: what the vehicle cost new, how old it is, how far it has been driven, its fuel type and who is selling it. This project applies Exploratory Data Analysis (EDA) to a real resale dataset of **301 listings and 9 variables** to discover which of these factors are genuinely associated with the selling price, and how strongly.

The specific goals are to:

- understand the structure, data types and quality of the dataset and document every cleaning decision;
- summarise numeric and categorical features with descriptive statistics;
- visualise distributions, comparisons, relationships and correlations using seven charts of six different types;
- turn the visual evidence into clear, data-backed findings, and state honestly what the data cannot show.

Approach and tools

The work follows a standard EDA flow: load and inspect, clean, summarise, visualise, interpret. Analysis was carried out in Python (pandas, NumPy, Matplotlib, Seaborn) on the project dataset *car data.csv*. The charts in this report were regenerated from the same file with Matplotlib so that they share one visual theme; every statistic quoted was recomputed from the data. Prices are in **INR lakh** (1 lakh = 100,000 rupees); the file does not state its unit, so this is an assumption based on the magnitude of the values.

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02 Dataset Overview

Source file: car data.csv

301
TOTAL RECORDS
as loaded from the CSV

9
COLUMNS
4 text + 5 numeric

0
MISSING VALUES
across all 2,709 cells

2
DUPLICATE ROWS
exact copies, removed

2003-2018
MODEL YEARS
median year 2014

198 / 101
CARS / TWO-WHEELERS
after cleaning (n=299)

3.51
MEDIAN SELLING PRICE
INR lakh, all vehicles

5.25
MEDIAN CAR PRICE
INR lakh, cars only

What the dataset contains

Each row is one used-vehicle resale listing. The target of interest is **Selling_Price**, the price at which the vehicle is being sold. The remaining columns describe the vehicle (age, original price, usage, fuel, gearbox) and the sale (seller type, number of previous owners). All 9 columns are fully populated.

Column	Type	Meaning	Example values
Car_Name	Text	Model name of the vehicle	ritz, city, fortuner, Bajaj Pulsar 150 (98 unique raw names)
Year	Integer	Year of manufacture	2003 to 2018
Selling_Price	Decimal	Price the vehicle is being sold at (INR lakh)	0.10 to 35.00
Present_Price	Decimal	Showroom (new) price of the model (INR lakh)	0.32 to 92.60
Kms_Driven	Integer	Distance already driven (km)	500 to 500,000
Fuel_Type	Category	Fuel used	Petrol, Diesel, CNG
Seller_Type	Category	Who is selling	Dealer, Individual
Transmission	Category	Gearbox type	Manual, Automatic
Owner	Integer	Number of previous owners	0, 1, 3

Column meanings are interpreted from the column names; the file does not ship a data dictionary.

Important discovery during profiling

Although the file is called a cars dataset, **101 of the 299 de-duplicated listings are motorcycles and scooters** (for example Bajaj Pulsar, Royal Enfield, Hero Splendor, Honda Activa). They sell for about one-tenth of a car, so mixing them in affects almost every summary statistic. This report therefore analyses the whole file where useful and **cars only** wherever the comparison would otherwise be distorted. Section 3 explains how the two groups were separated.

03 Data Cleaning & Preparation

Checks, actions and results

Every check was performed in pandas before any analysis. The table records what was checked, what was found and the action taken.

Check	Finding	Action taken
Missing values	0 nulls in every column (301 x 9 table)	No imputation required
Duplicate rows	2 exact duplicate rows (Ertiga 2016 and Fortuner 2015 each appear twice)	Removed with drop_duplicates: **301** → **299 rows**
Inconsistent text	11 Car_Name values with stray or double spaces (e.g. "KTM 390 Duke ", "Yamaha FZ v 2.0")	Whitespace trimmed and collapsed; unique names 98 → 97
Mixed vehicle types	101 rows are two-wheelers (Bajaj, Hero, Honda, KTM, Royal Enfield, TVS, Yamaha and others)	New column **Vehicle_Type** (Car / Two-wheeler), assigned by matching the 37 car model names in the data
Vehicle age	Year alone is hard to interpret directly	New column **Age = 2018** – **Year** (2018 is the latest year present in the data)
Resale retention	Prices span two orders of magnitude	New column **Retained = Selling / Present price** to compare depreciation fairly
Outliers	16 selling prices above the IQR fence of 13.73 lakh; one Present_Price of 92.60 (Land Cruiser); one Kms_Driven of 500,000 (Honda Activa 3g, 2008)	Retained: all are plausible real vehicles, not entry errors. Handled by using medians and a log scale
Encoding	Fuel, seller and gearbox are text categories	One-hot encoded with get_dummies (as in the notebook) for later modelling; charts use the original labels

Effect of cleaning

Statistic	Raw file (301 rows)	Cleaned (299 rows)
Mean Selling Price (INR lakh)	4.66	4.59
Median Selling Price (INR lakh)	3.60	3.51
Mean Kms Driven	36,947	36,917
Mean Present Price (INR lakh)	7.63	7.54

Reading the result

Removing two duplicate listings changes the averages only marginally, which shows the dataset was already in good shape. The larger analytical decision was separating cars from two-wheelers, because that single distinction explains most of the spread in price.

04 Descriptive Statistics

Cleaned data, n = 299

Variable	Mean	Std dev	Min	25%	Median	75%	Max
Year	2013.6	2.9	2003	2012	2014	2016	2018
Selling Price (lakh)	4.59	4.98	0.10	0.85	3.51	6.00	35.00
Present Price (lakh)	7.54	8.57	0.32	1.20	6.10	9.84	92.60
Kms Driven	36,917	39,015	500	15,000	32,000	48,884	500,000
Owner	0.04	0.25	0.00	0.00	0.00	0.00	3.00

Table 1. Summary of numeric variables (count = 299 for each).

Group	Listings	Median price	Median present price	Median Kms	Fuel mix
Cars	198	5.25	8.45	39,742	Petrol 138, Diesel 58, CNG 2
Two-wheelers	101	0.50	0.84	18,000	Petrol 101

Table 2. Cars versus two-wheelers (prices in INR lakh).

Categorical profile

79.9%
PETROL LISTINGS
239 of 299

87.0%
MANUAL GEARBOX
260 of 299

64.5%
DEALER SALES
193 of 299

96.3%
FIRST-OWNER
288 of 299 (Owner = 0)

Interpretation

- **Skewed prices.** The mean selling price (4.59) sits above the median (3.51) and the skewness is 2.54: most listings are cheap and a few expensive SUVs pull the average up. The median is the safer summary.
- **Wide range of usage.** Kms Driven has a median of 32,000 km but a maximum of 500,000 km and a standard deviation larger than its mean, so a few extreme values dominate its spread.
- **Recent stock.** Half of all listings were made between 2014 and 2018; the central 50% fall in 2012-2016.
- **Almost no variation in ownership.** 288 of 299 listings have Owner = 0, so this column carries very little information.
- **Fuel and CNG.** CNG has only 2 listings, too few to draw conclusions, so it is excluded from fuel comparisons.

05 Visual Analysis

Seven charts, six chart types

Univariate analysis: how are prices and categories distributed?

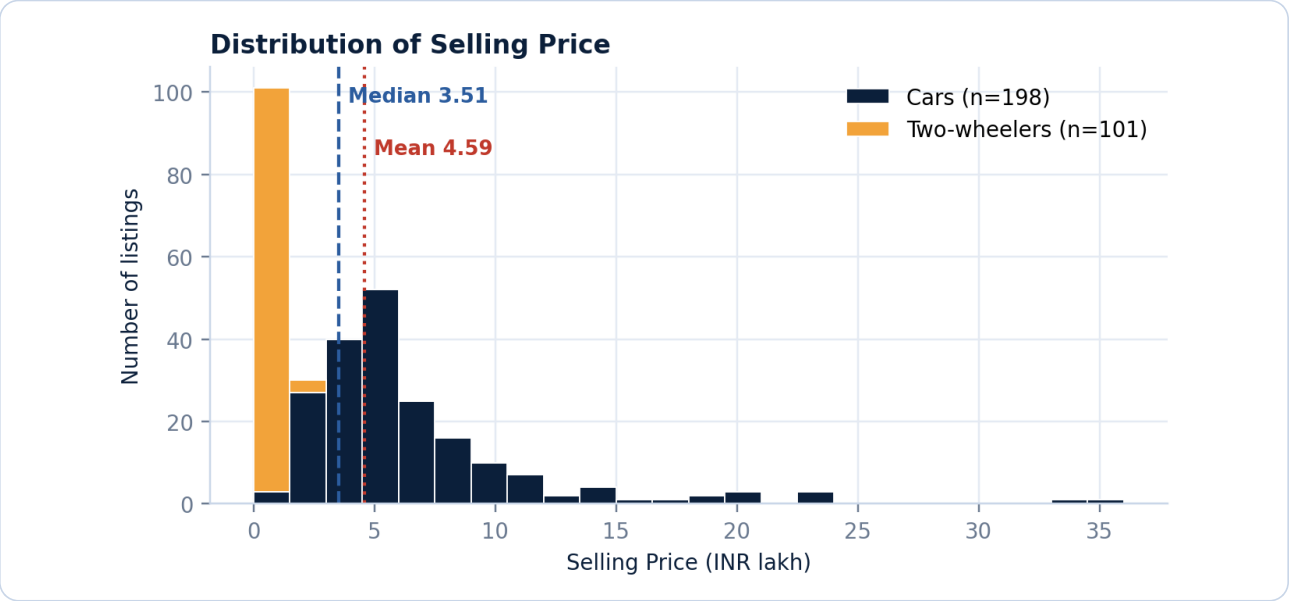


Figure 1. Histogram of Selling Price, stacked by vehicle type (all 299 listings).

Observation

The distribution is strongly right-skewed. All 101 two-wheelers fall in the lowest price band (under 1.5 lakh), while cars form a second hump around 3-6 lakh and trail off to 35 lakh. The mean (4.59) is pulled above the median (3.51) by a small group of expensive cars.

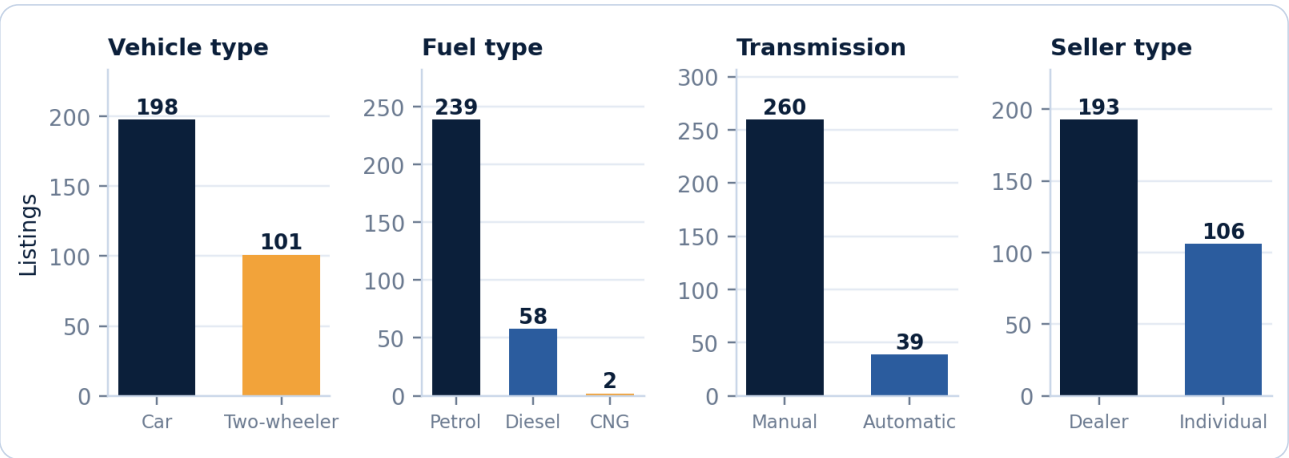


Figure 2. Bar charts of listing counts by vehicle type, fuel type, transmission and seller type.

Observation

The market is dominated by petrol (239), manual (260) vehicles. Diesel accounts for 58 listings, automatics for only 39 and CNG for 2. Dealers handle 193 listings and individuals 106, but that split is entangled with vehicle type (Finding 6).

Comparison and relationship analysis

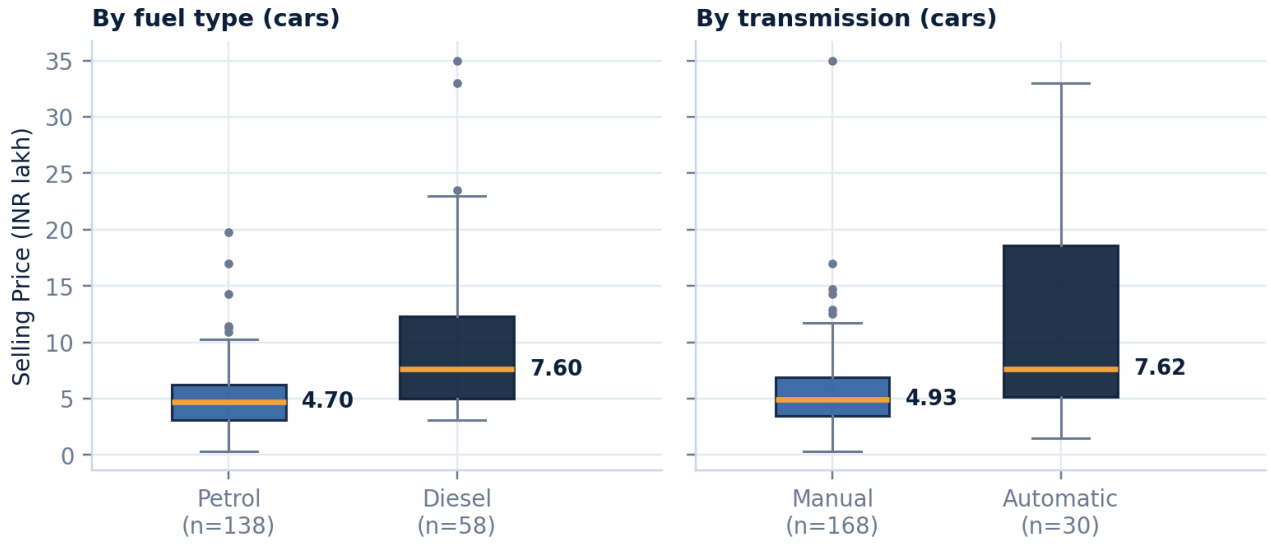


Figure 3. Box plots of car selling price by fuel type and by transmission (cars only; CNG excluded, n = 2).

Observation

Diesel cars have a median price of 7.60 lakh versus 4.70 for petrol, and automatic cars 7.62 versus 4.93 for manual. The automatic and diesel boxes are also much taller, meaning their prices vary far more. Part of the premium is a segment effect: these cars had higher showroom prices to begin with (Finding 4).

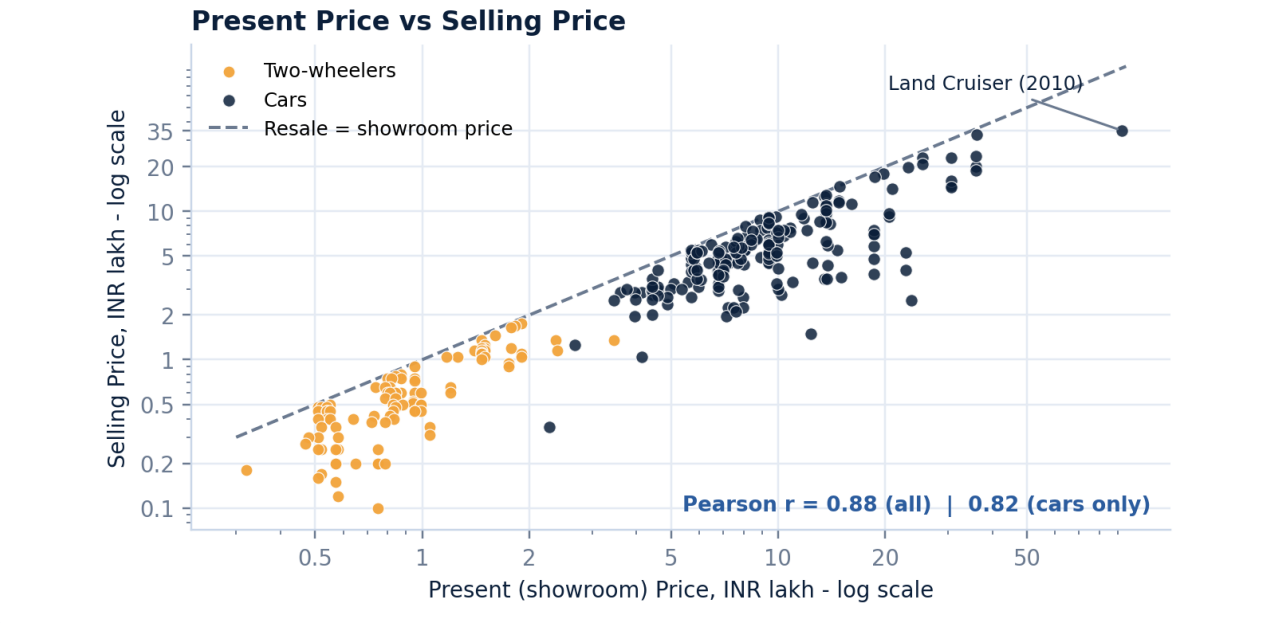


Figure 4. Scatter plot of Present Price against Selling Price on log axes, coloured by vehicle type.

Observation

The two prices move together almost in a straight line (r = 0.88 overall, 0.82 for cars). **Every one of the 299 listings sits on or below the "resale = showroom price" line**: the highest resale-to-new ratio is 0.99. Cars far below the line (bottom right) are expensive models that lost the most value.

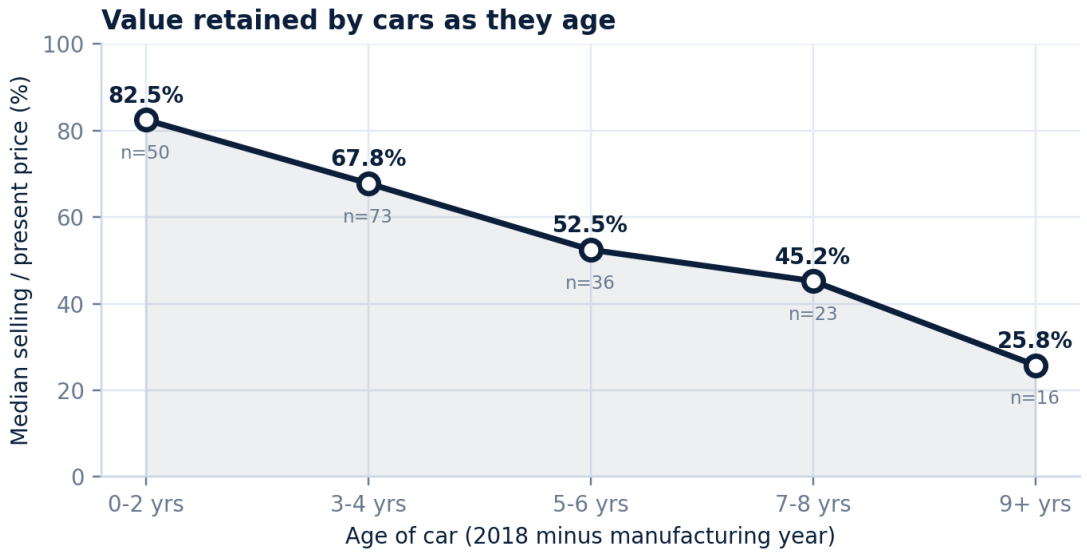


Figure 5. Line chart of the median share of showroom price retained, by age of car (cars only).

Observation

Resale value falls steadily with age: cars aged 0-2 years keep a median 82.5% of their showroom price, dropping to 52.5% at 5-6 years and just 25.8% beyond 8 years. The correlation between age and retained value is r = -0.86. The 9+ year group has the fewest cars (n = 16), so treat its exact value cautiously.

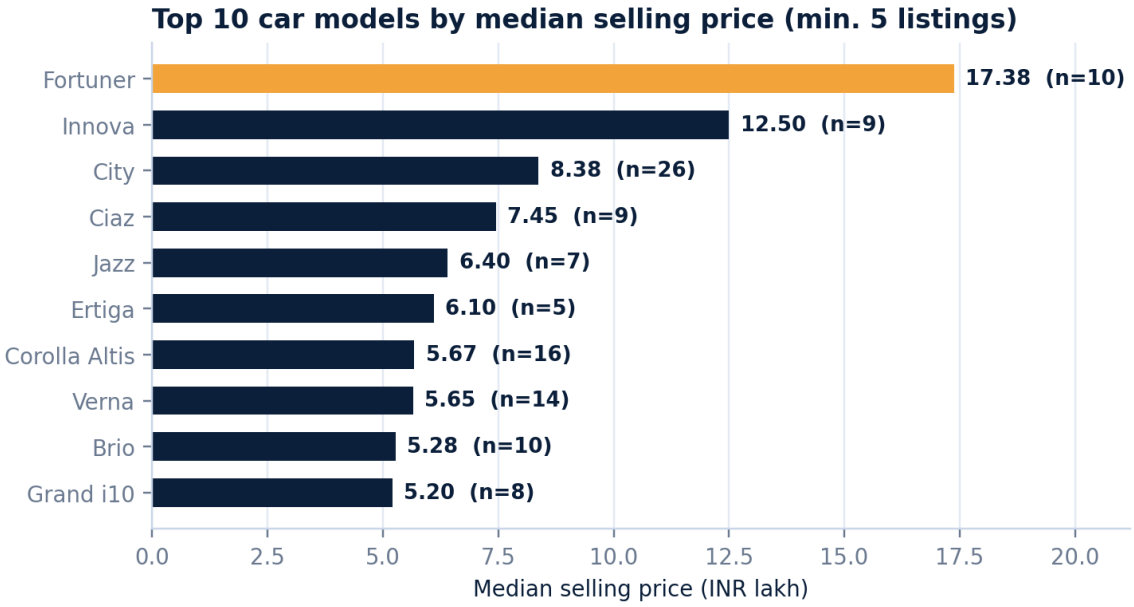


Figure 6. Horizontal bar chart of the ten highest-priced car models (models with at least 5 listings).

Observation

Toyota SUVs lead: Fortuner (17.38 lakh median) and Innova (12.50) are clearly above the rest. The Honda City is the most listed model (26 listings, median 8.38). Models such as Brio, Verna and Grand i10 cluster tightly between 5 and 6 lakh.

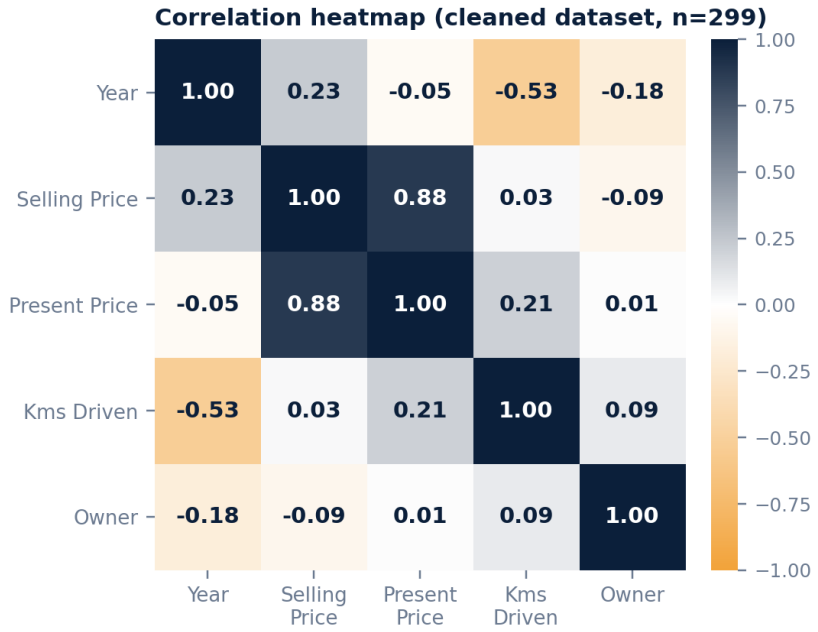


Figure 7. Correlation heatmap of the five numeric variables (Pearson r, cleaned data).

Observation

Present Price is by far the strongest partner of Selling Price (r = 0.88). **Year** has a modest positive link (0.23). **Kms Driven** shows almost none (0.03), yet it is strongly tied to Year (-0.53): newer vehicles have been driven less. Owner is weakly negative (-0.09).

How to read the heatmap

Values near +1 (navy) mean two variables rise together; values near -1 (amber) mean one rises as the other falls; values near 0 mean no linear relationship. The heatmap is computed on all 299 cleaned listings, as in the notebook. Because cars and two-wheelers are mixed, some coefficients differ from the cars-only values reported in the findings. For example, the Selling Price and Kms Driven correlation is 0.03 overall but -0.11 for cars alone, and Year versus Kms Driven is -0.53 overall but -0.71 for cars.

Correlation measures linear association only. It does not prove that one variable causes another.

Chart inventory

Figure	Chart type	Variables	Purpose
1	Histogram (stacked)	Selling Price by vehicle type	Shape and skew of the price distribution
2	Bar charts (4 panels)	Type, fuel, transmission, seller	Composition of the market
3	Box plots (2 panels)	Price by fuel and by transmission	Group comparison and spread
4	Scatter plot (log axes)	Present Price vs Selling Price	Strength of the price relationship
5	Line chart	Retained value by car age	Depreciation trend
6	Horizontal bar chart	Median price by model	Which models command the most
7	Heatmap	Five numeric variables	Correlation structure

Seven charts across six distinct chart types, all drawn from the cleaned dataset.

06 Key Findings

Seven evidence-based insights

01 — 10x PRICE GAP

Two very different vehicle markets share one file

101 of 299 listings (34%) are two-wheelers. Their median price is 0.50 lakh against 5.25 for cars, about ten times lower. Any analysis that ignores this mixes two different markets.

02 — r = 0.88

Showroom price is the strongest predictor of resale price

Present Price correlates with Selling Price at r = 0.88 (all) and 0.82 (cars). No listing sells above its showroom price; the best ratio is 0.99.

03 — 82% to 26%

Cars lose value quickly with age

Median retained value falls from 82.5% (0-2 years) to 25.8% (9+ years), with r = -0.86 between age and retained share.

04 — +2.9 LAKH DIESEL

Diesel and automatic cars sell for more, partly because they cost more new

Diesel median 7.60 vs petrol 4.70 lakh; automatic 7.62 vs manual 4.93. Their showroom medians are also higher (diesel 10.38 vs 7.60; automatic 17.41 vs 7.88), so the premium reflects model segment as much as fuel or gearbox.

05 — r = 0.03

Kilometres driven is a weak stand-alone predictor

Kms Driven correlates with Selling Price at only r = 0.03 (all) and -0.11 (cars). It is tangled with age (Year vs Kms r = -0.71 for cars), so its effect is largely already captured by Year.

06 — 101/101 INDIVIDUAL

Seller type cannot be compared fairly in this data

193 of 198 cars are sold by dealers, while all 101 two-wheelers are sold by individuals. Comparing dealer and individual prices would really compare cars with bikes, so no seller-type conclusion is drawn.

07 — 16 OUTLIERS

Prices are skewed by a handful of premium SUVs

Skewness is 2.54; 16 listings exceed the 13.73 lakh outlier fence and all are cars (Fortuner 8, Innova 4, Corolla Altis 3, Land Cruiser 1). Medians describe the market better than means.

07 Limitations

What this analysis cannot tell us

- **Small sample.** Only 299 listings after cleaning, and some groups are tiny (CNG n = 2; Owner = 3 has 1 listing; 9+ year cars n = 16).
- **Missing drivers of price.** The data has no information on city, condition, accident history, insurance, colour, engine size, or fuel economy, all of which affect resale value.
- **Two vehicle classes.** The dataset mixes cars and two-wheelers. The Vehicle_Type column was created by matching model names manually; it is reliable for this file but is not an original field.
- **Confounded variables.** Seller type is fully confounded with vehicle type, and kilometres with age, so single-variable comparisons must be read with care.
- **Assumed units and reference year.** Prices are assumed to be in INR lakh, and Age uses 2018 as the reference year because it is the newest model year present. A single snapshot also gives no information about price changes over time.
- **Association, not causation.** Correlations and group medians describe patterns; they do not prove that fuel type or transmission alone causes a price difference.

08 Conclusion & Future Scope

Conclusion

This EDA examined 301 used-vehicle listings, removed 2 duplicates, standardised vehicle names, and separated 198 cars from 101 two-wheelers. The evidence shows that a vehicle's resale price is driven primarily by **what it cost new** (r = 0.88) and by **how old it is**, with cars keeping about 82% of their showroom value in their first two years and only about 26% after nine. Diesel and automatic cars command higher prices, largely because they belong to pricier segments, while kilometres driven and ownership history add little on their own. The analysis also exposed a data-quality issue, the presence of two-wheelers and the dealer/individual overlap, that would have misled a model built on the raw file.

Future scope

- **Predictive modelling.** Train Linear Regression, Random Forest and Gradient Boosting models on Present Price, Age, fuel, transmission and vehicle type, and compare them with R-squared and RMSE.
- **Separate models** for cars and two-wheelers, and a log transform of price to handle the skew.
- **Richer data.** Collect more listings with city, condition, engine capacity and mileage-per-litre, and include CNG and electric vehicles.
- **Depreciation curves.** Estimate how quickly each model loses value and build a simple resale-price estimator or dashboard for buyers and sellers.

PROJECT NOTEBOOK

View the full Python code and outputs on Google Colab

Open notebook >

Link: https://colab.research.google.com/drive/1rSY7ee2D07S4x3uUuPyh1lu_jNZ0NAT7